FLATNESS DESCENDS FROM THE COMPLETION TO THE ORIGINAL LOCAL HOMOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness of a local homomorphism tested on the quotients by the powers of the maximal ideal. the author cites Algebra, Lemma algebra-lemma-flat-module-powers at proof lines 10755-10758. Converts the established flatness of the quotients into flatness of the map itself.
- Flatness after completion: a map is flat when its completion is flat. the statement counted by the sibling card labelled lemma-flat-after-completion of the same section, cited at proof lines 10748-10750. Supplies the reduction to the completed map.
- Identification of the quotients of the completion by powers of the maximal ideal with the quotients of the base. used at proof lines 10740-10752. Makes the hypothesis applicable to the family of quotients required by the criterion.

Source reference. Original lemma 0C4G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15226–15226. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0584: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6767–6767. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0523: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6556–6556. Navigation pointer to a statement the

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Lemma 0.2. *Let $A \to B$ be a local homomorphism of Noetherian local rings. Then the induced map of completions $A^\wedge \to B^\wedge$ is flat if and only if $A \to B$ is flat.*

Proof. Consider the commutative diagram

$$\begin{array}{ccc} A^\wedge & \longrightarrow & B^\wedge \\ \uparrow & & \uparrow \\ A & \longrightarrow & B \end{array}$$

The vertical arrows are faithfully flat. Assume that $A^\wedge \to B^\wedge$ is flat. Then $A \to B^\wedge$ is flat. Hence B is flat over A by Algebra, Lemma 0584.

Assume that $A \to B$ is flat. Then $A \to B^\wedge$ is flat. Hence $B^\wedge/\mathfrak{m}_A^n B^\wedge$ is flat over $A/\mathfrak{m}_A^n$ for all $n \geq 1$. Note that $\mathfrak{m}_A^n A^\wedge$ is the nth power of the maximal ideal $\mathfrak{m}_A^\wedge$ of $A^\wedge$ and $A/\mathfrak{m}_A^n = A^\wedge/(\mathfrak{m}_A^\wedge)^n$. Thus we see that $B^\wedge$ is flat over $A^\wedge$ by applying Algebra, Lemma 0523 (with $R = A^\wedge$, $I = \mathfrak{m}_A^\wedge$, $S = B^\wedge$, $M = S$). □